ETHNOHACK 2026 · TRACK 1 · TRADITIONAL KNOWLEDGE MINING

Was Pliny right?

Two-thousand-year-old plant remedies, checked against modern science

Team Parse-ian-track1

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THE QUESTION

Pliny recorded hundreds of plant remedies. Does modern science agree?

Old texts hide their knowledge behind old names: plants like *orage* or *hibiscum*, diseases like *dropsy* or *quinsy*. To compare them with modern research, both must first be translated, remedy by remedy.

1,720
remedies counted by Pliny himself

100
chapters of *Natural History*, Book 20 (1855 translation)

548
footnotes by 19th-century editors

THE PIPELINE

Six steps from Pliny’s sentence to a modern verdict

1

Text

Chapters, sentences and footnotes from the 1855 edition

2

Claims

A local language model extracts every remedy with its exact quote

3

Plants

Footnotes, the model and World Flora Online name each plant

4

Conditions

Old terms mapped to medical vocabulary (MeSH)

5

Evidence

Europe PMC searched for every plant and condition

6

Graph

Knowledge graph and an explorer for every remedy

Every result traces back to Pliny’s exact words, and the team reviewed the critical decisions at every stage.

TRANSLATING THE PAST

Old names, modern meaning

Pliny wrote	Today	Decided by
orage	Atriplex hortensis (garden orache)	1855 footnote, confirmed in World Flora Online
hibiscum	Althaea officinalis (marsh mallow)	team review of the text
dropsy	Edema	medical vocabulary (MeSH)
quinsy	Peritonsillar abscess	medical vocabulary (MeSH)

99.8% of claims linked to a modern plant

94% of conditions mapped to modern medicine

QUALITY

Measured at every step, checked by hand where it matters

0.82

F1 score of claim extraction on held-out chapters never seen during development

45

plant and condition identifications reviewed and decided by the team

56

evidence verdicts checked by hand against the papers behind them

The language model proposes; curated vocabularies, the 1855 editors and the team decide.

WHAT MODERN SCIENCE SAYS

Half of Pliny's remedies have never been studied.

51%
no paper on the plant and the condition at all

31%
papers exist, but none tests the remedy

≤14%
partly studied, in animals or the lab

5
remedies confirmed by human studies

CONFIRMED

Five of Pliny’s remedies hold up in modern human studies

Onion juice for hair loss	Clinical trial in alopecia areata
Anise for indigestion	Two randomised trials in functional dyspepsia
Marsh mallow for cough	Review of animal and clinical studies
Mustard for deep-seated pain	Trial of a mustard plaster in knee osteoarthritis
Black cumin for mouth ulcers	Trial in erosive oral lichen planus

All 32 remedies the pipeline marked as supported were checked by hand: 5 verified, 9 with indirect support, 18 rejected.

WHO SAID IT

Ancient authority pointed to the right questions, not the right answers

Share of claims	Named physicians	Pliny himself	Difference
Studied, at least in the lab	20%	13%	significant (p ≈ 0.01)
Never studied	44%	52%	significant (p ≈ 0.03)
Supported in humans	2%	3%	no difference (p ≈ 0.5)

203 claims Pliny attributes to named physicians such as Hippocrates, 1,210 in his own voice; hearsay (112) does not differ. Exploratory.

LEADS

307 untested remedies worth a closer look

Rue for insect stings	7 claims, 2 sources
Marsh mallow for insect stings	4 claims
Mallow for insect stings	4 claims
Purslane as an antidote	3 claims
Radish against scorpion stings	2 claims

Explore every remedy

A knowledge graph of 112 plants, 351 conditions, 1,525 claims and 253 papers. Pick a plant, see Pliny’s words, the verdict and the modern papers.

https://ailrezashafaee.github.io/Ethnohack/data/06_graph/explorer.html

LIMITS AND NEXT STEPS

A map of where ancient knowledge and modern evidence meet

What to keep in mind

- Automated support needs a human check: 5 of 32 held up
- “Untested” means not found by our search, not proven wrong
- Some plant names remain uncertain

What comes next

- Stricter reading of the evidence, and EU herbal monographs
- More books of Pliny, other authors such as Dioscorides
- Ranking the 307 leads by their known compounds

Historical claims, not medical advice. Code, data and results: github.com/ailrezashafaee/Ethnohack